Brieuc Collet, Researcher in Space Physics

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Space physics researcher specialized in planetary wave-plasma instability

Employment History

2025– 2027 **Post Doc** PPARC, Tohoku University, Japan
Understanding Mercury and the Jovian environment with in situ radio measurements from Juno, BepiColombo and Juice

2022– 2025 **PhD in Astrophysics** Lab of Astrophysics of Marseille, Aix-Marseille University, France
Understanding the microphysics of Jovian radio auroral emissions with Juno space probe

Education

2022 – 2025 **PhD in Astrophysics, Aix-Marseille Université**

2021 – 2022 **Master's Degree Plasma Physics, Paris Saclay University**

2019–2020 **Bachelor's Degree of Fundamental Physics, Paris Saclay University**

Teachings

2022-2025 **Teaching Assistant**, 192h, Aix-Marseille University

Scientific Outreach

2022–2025 **Festival of Astronomy of Provence**

Fête de la Science - French Science Diffusion Day

Skills

Programming Proficiency in **Python**, Knowledge in **C++** and **IDL**

Astronomy Observation nights at Observatory of Haute Provence

Community Activities

2022–2025 **Laboratory's Sustainable Development Group Member**

Laboratory's Seminar Organizing Committee

2023–2025 **PhD Representative in Laboratory Council**

2025 **Local Organizing Committee** for PREX, Marseille France

2025-2026 **Reviewer** for PREX proceedings and Geophysical Research Letters

Associate Scientist BepiColombo/Mio/PWI/Sorbet

2026 **Metadata creation for VESPA** for Hisaki and IPRT data

Awards

2022 PhD grant

2025 **Best PhD presentation award**, Planetary, solar and heliospheric Radio Emissions X, Marseille, France

Publications

[1] Al Saati, S., Clément, N., […], **Collet, B.**, et al. 2022. "Magnetosphere -Ionosphere -Thermosphere Coupling Study at Jupiter Based on Juno's First 30 Orbits and Modeling Tools". In: *Journal of Geophysical Research: Space Physics*. DOI: 10.1029/2022JA030586.

[2] Benmahi, B., Hue, V., […], **Collet, B.**, et al. 2025. "Precipitating electron properties of Ganymede's aurora retrieved from Juno/UVS observations during PJ34". In: *Astronomy &; Astrophysics*. DOI: 10.1051/0004-6361/202556996.

[3] **Collet, B.** 2025. "Understanding the microphysics of Jovian radio auroral emissions with Juno". PhD thesis supervised by Lamy, L. at Laboratoiry of Astrophysics of Marseille, Aix-Marseille University. PhD thesis.

[4] **Collet, B.**, Lamy, L., Kasaba, Y., and Andre, N. 2026. "Reanalysis of SKR Source Crossed by Cassini: Loss Cone Electrons as a Secondary Source of Free Energy". In: *Geophysical Research Letters* 53.16. ISSN: 1944-8007. DOI: 10.1029/2026gl125087.

[5] **Collet, B.**, Lamy, L., et al. 2023. "Characterization of Jovian hectometric sources with Juno: statistical position and generation by shell-type electrons". In: *PRE IX*. Ed. by C. K. Louis, C. M. Jackman, et al. DIAS and TCD. DOI: 10.25546/103095.

[6] **Collet, B.**, Lamy, L., et al. 2024. "A New Type of Jovian Hectometric Radiation Powered by Monoenergetic Electron Beams". In: *Journal of Geophysical Research: Space Physics*. DOI: 10.1029/2024JA032422.

[7] **Collet, B.**, Lamy, L., et al. 2025. "In Situ Analysis of Jupiter's Broadband Kilometric Auroral Radio Emissions With Juno". In: *Geophysical Research Letters* 52.9. DOI: 10.1029/2024gl114447.

[8] **Collet, B.**, Lamy, L., et al. 2026. "Detection of Jovian kilometric and hectometric auroral radio harmonics with Juno/Waves in situ measurements". In: *Planetary, Solar and Heliospheric Radio Emissions X*. Ed. by L. Lamy, C. K. Louis, et al. Accepted, preprint. OSU Pythéas/AMU, Observatoire de Paris. DOI: 10.25935/PREX-TR41.

[9] Heidari, N., Boisse, I., […], **Collet, B.**, et al. 2024. "The SOPHIE search for northern extrasolar planets: XIX. A system including a cold sub-Neptune potentially transiting a V = 6.5 star HD 88986". In: *Astronomy &; Astrophysics*. DOI: 10.1051/0004-6361/202347897.

[10] Lamy, L., Cecconi, B., […], **Collet, B.**, et al. 2023. "Re-exploring the radio spectrum of Uranus in orbit: science case and digital high-frequency receiver". In: *PRE IX*. Ed. by C. K. Louis, C. M. Jackman, et al. DIAS and TCD. DOI: 10.25546/103105.

[11] Louis, C. K., Boudouma, A., […], **Collet, B.**, et al. 2026. "Search for the influence of the Galilean moons on Jovian kilometric and decametric radio emissions over 7 years of Juno data". In: *Planetary, Solar and Heliospheric Radio Emissions X*. Ed. by L. Lamy, C. K. Louis, et al. OSU Pythéas/AMU, Observatoire de Paris. DOI: 10.25935/PREX-VWFQ.

[12] Louis, C. K., Louarn, P., **Collet, B.**, et al. 2023. "Source of Radio Emissions Induced by the Galilean Moons Io, Europa and Ganymede: In Situ Measurements by Juno". In: *Journal of Geophysical Research: Space Physics*. DOI: 10.1029/2023JA031985.

[13] Rabia, J., Hue, V., […], **Collet, B.**, et al. 2025. "In situ and remote observations of the ultraviolet footprint of the moon Callisto by the Juno spacecraft". In: *Nature Communications*. DOI: 10.1038/s41467-025-62520-4.

[14] Wu, Siyuan, Whiter, Daniel K., […], **Collet, Brieuc**, et al. 2026. "A radio oval above Earth's auroral oval". In: *Science Advances*. DOI: 10.1126/sciadv.aec4114.

Conferences

2022 **French National Sun-Planet meeting (ATST)**, Marseille, France

Planetary, solar and heliospheric Radio Emissions IX, Dublin, Ireland

2023 **Chapman Understanding Alfvén Waves in the Sun and the Heliosphere**, Berlin, Germany

2024 **Astroradiofr24, French radio frequency astrophysics, towards SKA**, Paris

French National Sun-Planet meeting (ATST), Marseille, France

Magnetospheres of Outer Planets, Minneapolis, USA

2025 **EGU General Assembly**, Vienne, Austria

Planetary, solar and heliospheric Radio Emissions X, Marseille, France

2026 **Symposium on Planetary Sciences**, Sendai, Japan

JPGU, Chiba, Japan (next May)

French National Sun-Planet meeting (ATST), Biarritz, France

Magnetospheres of Outer Planets, Toulouse, France (next July)